The Cellular Concept

Objectives of Wireless Systems

- Large Capacity
- Efficient use of Resources (Spectrum)
- Availability
- Adaptability to traffic density
- Service to mobile terminals
- Quality of Service
- Affordability

Introduction of Cellular Concept

- The need to operate and grow “indefinitely” within an allocation of hundred of channels has been the primary driving force behind the evolution of the cellular concept.

- The concept was developed in 1947 at AT&T Bell Laboratories. First tests were conducted in 1962 for commercial applications. FCC finally set aside new radio frequencies for land mobile communications in 1970. In 1970, AT&T proposed to build the first high capacity cellular phone system called “Advanced Mobile Phone Service (AMPS)”.

Evolution of Cellular Networks

- 1st Generation Cellular Networks: High power, high tower base stations, analog systems, FDMA. “The probability that a mobile will leave the coverage area during the course of a call was considered negligible”. Wireless was nothing but a mode of access to the fixed telecommunication network.

- 2nd Generation Cellular Networks: Digital systems… TDMA in North America (IS 54) and Europe (GSM) CDMA (IS-95) in North America

- 3rd generation: High bandwidth, service integration, high data rate

Basic Elements

- **Channel (Frequency) Reuse:**
  - Refers to the intelligent allocation and reuse of channels throughout different coverage areas which are separated from one another by sufficient distances so that co-channel interference is not objectionable.

- **Co-channel interference:**
  - Interference caused by transmissions on the same channel (e.g. same frequency)

- **Reuse distance**:
  - minimum distance between two cells using same channel for satisfactory signal to noise ratio

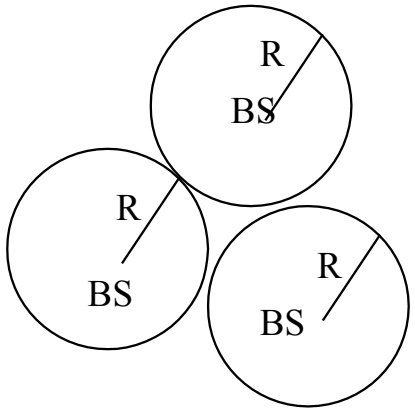
- Instead of covering the whole area with one transmitter (base station) of high power we deploy multiple base stations of moderate (lower) power. Each base station (cite) covers some specific area.

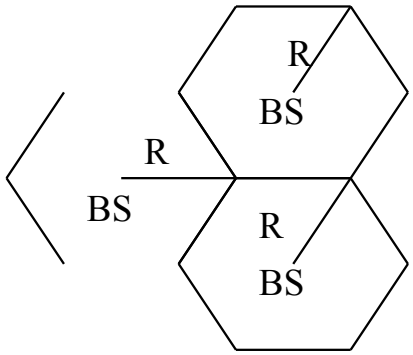
- Each base station is assigned a portion of the total number of channels, while neighboring base stations are assigned different groups of channels so that the interference between base stations (and mobile users) is minimized.

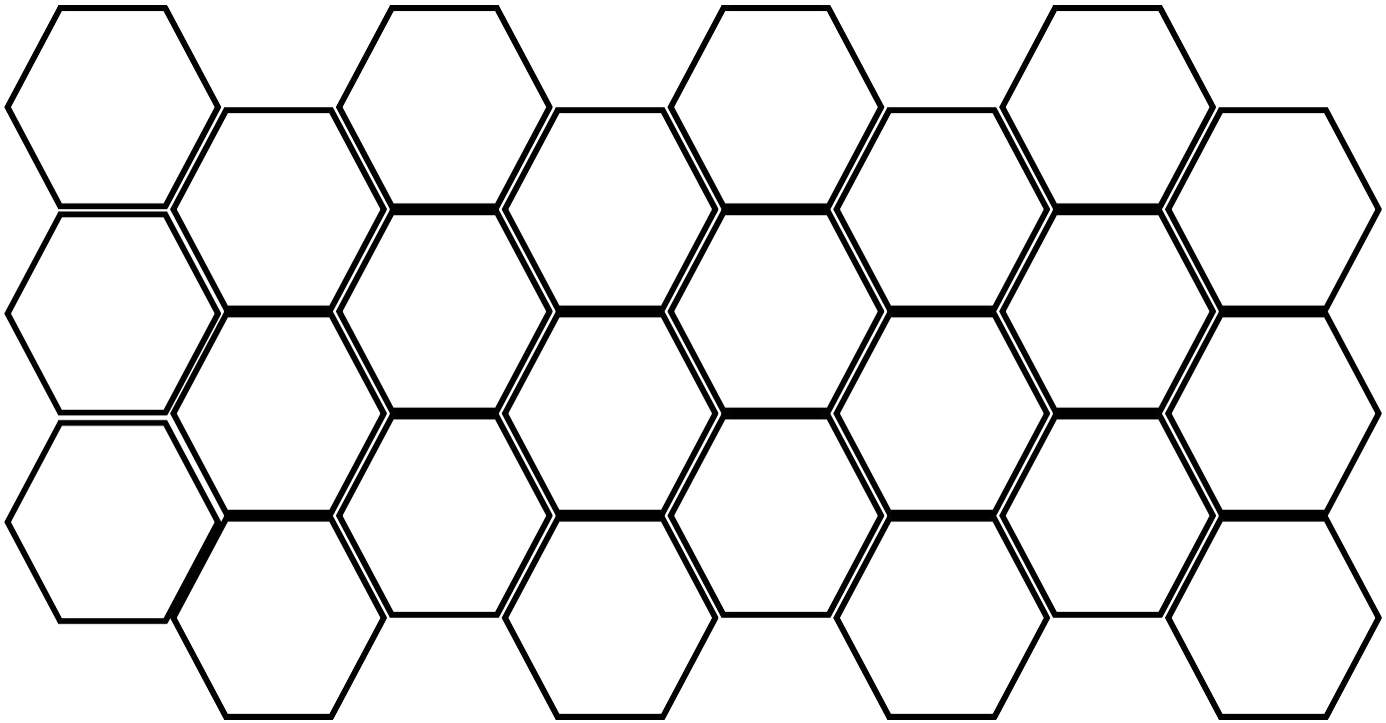
- Spacing the base stations systematically and allocating the channels appropriately results in minimizing the co-channel interference.

Hexagonal Structure

- Hexagons are usually used to depict the cells due to geometry considerations and calculation purposes (it closely approximates the circle, which is used as a coverage area by a base station that has transmission radius (range) R).







- Consider that there are S channels available (total in the system). If each cell is allocated K channels (K<S) and if S is divided among N cells into unique and disjoint channel groups which have the same number of channels, then we have:
- S = K * N
- The N cells that collectively use all the available channels are called clusters, and N is the cluster size. Typical values of N is 4,7, 12.
- If a cluster is replicated M times within the system then the total number of users supported is given by:
- C = M * K * N = M * S

- $C = M * K * N = M * S$

- For fixed service area if N decreases, while the cell size remains constant, more clusters are required to cover the area, and as a result M increases and therefore C increases.
- From a design point of view the smallest number of N is desired, so that M and thus C is maximized. However N represents how close cells that use the same channels are. Therefore smaller N corresponds to higher co-channel interference, while larger N corresponds to lower co-channel interference.
- Therefore minimum value of N should be obtained (desired), such that the co-channel interference can be tolerated. The value N is called reuse factor. (Some books define 1/N as the reuse factor).